

Course Title: **Digital Electronics and Logic Design**

Course Number: **CoSc2041**

Credit Hour: **3**

Prerequisite: **none**

COURSE DESCRIPTION:

This course deals with the fundamentals of digital systems building blocks. It starts from the basics of digital electronics – ICs and logic families. It continues discussing digital logic circuit elements like logic gates, Boolean algebra, Boolean function simplification, digital logic design like half and full adders, MSI chip elements like de-multiplexers and multiplexers, decoder and encoder. Moreover, students will be exposed to both combinational and sequential digital circuits where they analyze and design memory elements using MSI chips. Finally, the course will discuss digital logic families. Most of these concepts are supported by laboratory session to experiment the theory.

COURSE OBJECTIVES

At the end of the course students should be able to:

- Discusses the building blocks of digital electronics
- Explain the basic operation of various digital systems.
- Describe and solve problems involving digital systems using Boolean algebra or logic gates.
- Understand basic digital electronic systems.
- Analyze, design and simplify digital logic circuits using the various simplification and design techniques.

Course Content

Chapter 1: Introduction to digital computer systems

- 1.1. Digital computers and systems
- 1.2. Number Systems
- 1.3. Number systems conversion and
- 1.4. Arithmetic operations

Chapter 2: Digital ICs and Logic Families

- 2.1. Introduction to ICs and Logic Families
- 2.2. Digital IC Logic family's classifications
- 2.3. Evaluation of logic families
- 2.4. Characteristics of TTL gates

Chapter 3: Boolean algebra, switching algebra and logic gates

- 3.1. Boolean algebra basics
- 3.2. Truth table representation
- 3.3. Basic theorems, laws and rules of Boolean algebra
- 3.4. Boolean functions
- 3.5. Digital logic gates
 - 3.5.1. Basic logic gates
 - 3.5.2. Derived logic gates
- 3.6. Implementation of Boolean expression using logic gates

Chapter 4: Standard Boolean Expressions and Simplification

- 4.1. Need for Boolean Expression
 - 4.1.1. Why Standardization is needed?
 - 4.1.2. Types of Standard Boolean Expression
 - 4.1.2.1. Sum-Of –Product (SOP)
 - 4.1.2.2. Product-Of-Sum (POS)

4.1.3. NAND and NOR implementations

4.1.4. Why simplification is needed?

4.2. Methods of Simplifications

4.2.1. Algebraic Method

4.2.2. K-Map method

4.2.3. Quine-McCluskey method or Tabular Method

4.3. Don't care conditions

Chapter 5: Combinational Logic Circuits

5.1. Introduction

5.2. Design and Analysis Procedures

5.3. Arithmetic Circuit

5.3.1. Adders

5.3.2. Subtractors

5.3.3. Designing Adder- Subtractor Circuit

5.3.4. Multiplier

5.4. Multiplexer (MUXs)

5.5. Demultiplexer

5.6. Decoders

5.7. Encoders

5.8. Circuits for Code conversion

5.8.1. BCD to Excess-3 code

5.8.2. Excess-3 to BCD code

5.8.3. Gray code to Binary code

5.8.4. Binary to Gray code

5.8.5. BCD to 7-Segment

5.9. Decimal adder

5.10. Comparators

5.10.1. Designing 2-bits comparator

5.10.2. Designing 3-bits comparator

5.11. Seven Segment Display

Chapter 6: Sequential Logic Circuits

6.1 Introduction

6.2 Flip-Flops and Latches

6.3 Triggering of flip flops

6.4 Analysis of clocked sequential circuits

6.5 Flip-Flop Excitation table

6.6 Design with state equations

6.7 Memory unit Design

6.7.1. Read Only Memory (ROM) design

6.7.2. Programmable Logic Array (PLA) design

6.7.3. Programmable Array Logic (PAL) design

6.8. Registers and Counters

Chapter 7: Computer Aided Design Tools for Digital Circuits

7.1. Introduction to VDLs

Delivery Methods:

- Lectures
- Demonstrations and Discussions
- Reading Assignments

Methods of Assessment

Assessment Method	Weight
Written tests	20
Group project	20
Assignments	10
Final exam	50

Textbook

1. Morris M. Mano et. el., Digital Design,5th ed, Pearson Inc.
2. Thomas L. Floyd, Digital Fundamentals, 6th Ed., Prentice Hall

References

- Morris M. Mano, Computer System Architecture, 3rd Ed, Prentice Hall ,2015
- Roger L. Tokheim, Digital Electronics: Principles and Applications, McGraw Hill, 8th edition, McGraw hill ,2014
- Frederick F. Driscoll and Robert Coughlin, Operational Amplifiers and Linear ICs, 6th ed, Prentice Hall Inc, 2018
- Dueck, Robert. Digital Design with CPLD Application and VHDL.